

# MD RAKIBUL ISLAM RAIHAN

## ML Systems Engineer | LLM Inference & Multimodal AI

2027 New Grad · Available July 2027 · Open to Relocation · Xi'an, China

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## SUMMARY

ML Systems Engineer focused on LLM inference, GPU optimization, and multimodal generative AI. At InfiX.ai, optimized production LLM serving - Qwen on 4× A800 and DeepSeek on multi-node H200 clusters - reducing long-context TTFT 8.3× (55s→6.7s) via 3-tier KV-cache offload and raising stable concurrency 192→256 via RDMA-based prefill/decode disaggregation. First author of an IEEE TNNLS manuscript on NVFP4 4-bit audio-guided video diffusion (4× memory reduction on A800/H100). Strong in SGLang, CUDA/Triton, low-precision inference, and distributed systems.

## EXPERIENCE

### AI Infrastructure Engineer Intern, InfiX.ai, Shenzhen, China, Apr 2026 - Jun 2026

- Reduced long-context TTFT 8.3× (55s→6.7s) and sustained a **92%** cache hit rate by designing and deploying a 3-tier KV-cache offload system (GPU HBM → CPU DRAM → SSD) over Mooncake RDMA for Qwen3.5/3.6-27B on 4× A800-80GB (32K+ token contexts).
- Improved stable serving concurrency 33% (192→256 requests) by implementing 2+2 GPU prefill/decode disaggregation with RDMA-based KV transfer, RadixAttention prefix caching, and continuous batching for Qwen3.6-27B.
- Modified **SGLang** cache-management and memory-allocation internals (kvcacheio.py, memory\_pool\_host.py, hi\_mamba\_radix\_cache.py) to resolve production cache failures and support hybrid-attention (Mamba + full-attention) Qwen models.
- Profiled CUDA/Triton inference with **Nsight Systems** - GEMM accounted for **60%** of step time, redirecting optimization from cache I/O to compute kernels; FP8 KV cache halved per-token KV memory (72.9 → 36.5 KB). Deployed Qwen3-235B-A22B-FP8 (235B MoE) on 2× H200 and benchmarked DeepSeek-V3.2 (FP8) on 8× H200: **91 ms TTFT / 36 ms TPOT** at 32-64 concurrency, **13.16 req/s** at 256 concurrency.
- Owned root-cause analysis for **12+** production incidents across RDMA, etcd, storage, and distributed inference, restoring multi-node GPU-cluster reliability.

### Software Engineer Intern (AI Agent), Hong Kong Amram Group, Shenzhen, China, Jun 2025 - Sep 2025

- Developed production full-stack applications using React, REST APIs, and backend automation services.
- Built AI-agent workflow automation systems that improved operational efficiency by **25%**.

### Electrical Software Engineer Intern, Shaanxi Longong Intelligent Technology, Xi'an, China, Feb 2025 - Apr 2025

- Built CODESYS PLC industrial-control systems with Modbus/CAN/Ethernet-IP protocols, improving motor-control performance by **15%**.

## RESEARCH

### Graduate Researcher, Northwestern Polytechnical University, Sep 2024 - Present

Advisor: Prof. Zhang Peng

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer** achieving **4x memory reduction** while maintaining generation quality.
- Implemented low-precision quantization (**FP8/FP4/NVFP4**) for multimodal diffusion transformers; evaluated quality-efficiency trade-offs.
- Benchmarked NVFP4-DiT on **A800/H100 GPUs**; profiled end-to-end latency and memory with PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

### Undergraduate Researcher, Northwestern Polytechnical University, Jan 2023 - Jul 2024

Advisor: Dr. Yaxing Chen

- Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, achieving **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

## PROJECTS

### AI Infrastructure Portfolio, [github.com/theraihanrakibb/ai-infra-portfolio](https://github.com/theraihanrakibb/ai-infra-portfolio)

- Designed and shipped 10 Dockerized AI-infra services - LLM gateway, model serving, GPU observability, KV-cache simulation, benchmarking, quantization, tracing, and RAG - each with CI and a reproducible quickstart.

### Distributed Training Launcher, [github.com/theraihanrakibb/train-launcher](https://github.com/theraihanrakibb/train-launcher)

- Built a PyTorch distributed CLI that launches and supervises multi-node training jobs with fault detection, automatic retry, and checkpoint recovery.

## TECHNICAL SKILLS

**LLM Inference & Systems:** SGLang, vLLM, KV Cache, PD Disaggregation, Mooncake/RDMA, Continuous Batching, RadixAttention

**GPU & Performance:** CUDA, CUDA C++, Triton, Nsight Systems, Nsight Compute, GEMM optimization, FP8/FP4/INT4

**ML & Multimodal:** PyTorch, Hugging Face Transformers, Diffusion/DiT, Multimodal Learning, OpenCV

**Distributed Systems:** PyTorch Distributed, Docker, Kubernetes, Linux, Prometheus, CI/CD

**Programming:** Python, C++, CUDA C++, Bash, SQL, Data Structures & Algorithms

## EDUCATION

**M.Eng. in Software Engineering**, School of Software, Northwestern Polytechnical University - Sep 2024 - Jul 2027

GPA: 88/100, Top 1%, Chinese Government Scholarship, President of International Student Union (ISU)

**B.Eng. in Computer Science & Technology**, School of Computer Science, Northwestern Polytechnical University - Sep 2020 - Jul 2024

GPA: 85/100, Top 1%, Presidential Scholarship, Best Undergraduate Thesis Award

## PUBLICATIONS & HONORS

**First Author** — "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" — IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), manuscript under review.

**Honors:** Chinese Government Scholarship (2024-2027); Presidential Scholarship (2020-2024); Best Undergraduate Thesis Award (2024); Outstanding Graduate (2024), NWPU; Hackathon Winner - NWPU AI Challenge (2023).

# MD RAKIBUL ISLAM RAIHAN

## ML 系统工程师 | 大模型推理与多模态 AI

2027 届应届 · 2027 年 7 月到岗 · 可接受异地 · 中国·西安

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### 个人简历

ML 系统工程师, 专注大模型推理、GPU 优化与多模态生成式 AI。在 InfiX.ai 实习期间, 优化生产级大模型服务——Qwen 部署于 4× A800、DeepSeek 部署于 H200 多节点集群——通过三级 KV-cache 卸载将长上下文 TTFT 降低 8.3× (55s→6.7s), 并借助基于 RDMA 的 prefill/decode 分离将稳定并发由 192→256 提升。作为第一作者撰写 IEEE TNNLS 论文, 研究 NVFP4 4 比特音频引导视频扩散, 在 A800/H100 上实现显存降低 4×。擅长 SGLang、CUDA/Triton、低精度推理与分布式系统。

### 工作经历

#### 无界智索 (深圳) 科技 (InfiX.ai), AI 基础设施工程师实习, 深圳, 2026年4月 - 2026年6月

- 在 4× A800-80GB 上基于 Mooncake RDMA 设计并实现三级 KV-cache 卸载系统 (GPU HBM → CPU DRAM → SSD), 服务 Qwen3.5/3.6-27B 时将长上下文 (32K+ token) 首字延迟 (TTFT) 降低 8.3× (55s→6.7s), 缓存命中率维持 92%。
- 通过基于 RDMA 的 KV 传输实现 2+2 GPU prefill/decode 分离, 结合 RadixAttention 前缀缓存与连续批处理, 将 Qwen3.6-27B 的稳定服务并发提升 33% (192→256 请求)。
- 修改 SGLang 缓存管理与内存分配内部模块 (kvcacheio.py、memory\_pool\_host.py、hi\_mamba\_radix\_cache.py), 修复生产缓存故障并支持混合注意力 (Mamba + 全注意力) Qwen 模型。
- 使用 Nsight Systems 分析 CUDA/Triton 推理——GEMM 占单步耗时 60%, 由此将优化重心从缓存 I/O 转向计算算子; FP8 KV cache 将单 token KV 显存减半 (72.9 → 36.5 KB)。在 2× H200 部署 Qwen3-235B-A22B-FP8 (235B MoE), 并在 8× H200 上对 DeepSeek-V3.2 (FP8) 基准测试: 91 ms TTFT / 36 ms TPOT (32-64 并发)、13.16 req/s (256 并发)。
- 主导 12+ 起生产故障的根因分析与修复 (RDMA、etcd、存储与分布式推理), 恢复多机 GPU 集群可靠性。

#### 香港阿姆兰集团 (Amram Group), 软件工程师实习 (AI Agent), 深圳, 2025年6月 - 2025年9月

- 基于 React、REST API 与后端自动化服务开发生产级全栈应用。
- 开发 AI Agent 工作流自动化系统, 运营效率提升 25%。

#### 陕西龙工智能科技 (Longong), 电气软件工程师实习, 西安, 2025年2月 - 2025年4月

- 基于 CODESYS 开发 PLC 工业控制系统, 实现 Modbus、CAN 与 Ethernet/IP 通信协议, 电机控制性能提升 15%。

### 研究经历

#### 研究生研究员, 西北工业大学, 2024年9月 - 至今

导师: 张蓬 教授

- 开发 NVFP4-DiT: 4 比特音频引导视频扩散 Transformer, 显存降低 4 倍且保持生成质量。
- 实现多模态扩散 Transformer 的低精度量化 (FP8/FP4/NVFP4), 评估质量-效率权衡。
- 在 A800/H100 上对 NVFP4-DiT 进行基准测试, 使用 PyTorch Profiler 与 Nsight 分析端到端延迟与显存, 指导低精度算子选型。
- 基于视觉、音频与时序一致性的多模态深度伪造检测研究。
- 硕士论文: 《检测深度伪造视频: 基于多模态音视频框架与时序不一致性》。

#### 本科研究员, 西北工业大学, 2023年1月 - 2024年7月

导师: 陈亚星 博士

- 基于 Apache Spark 与 CKKS 同态加密设计并实现分布式机密查询框架, 通过优化分析管线实现 95% 隐私保护与 20% 性能提升。
- 本科论文: 《面向 Apache Spark 的分布式机密查询协议设计与实现》。

### 项目

#### AI 基础设施代码集, github.com/theraihanrakibb/ai-infra-portfolio

- 设计并交付 10 个 Docker 化的 AI 基础设施服务——LLM 网关、模型服务、GPU 可观测性、KV-cache 模拟、基准测试、量化、链路追踪与 RAG, 均配备 CI 与可复现的上手流程。

#### 分布式训练启动器, github.com/theraihanrakibb/train-launcher

- 开发 PyTorch 分布式命令行工具, 支持多节点训练任务的启动与监控, 并具备故障检测、自动重试与断点恢复。

### 技术技能

大模型推理与系统: SGLang, vLLM, KV Cache, PD 分离, Mooncake/RDMA, 连续批处理, RadixAttention

GPU 与性能: CUDA, CUDA C++, Triton, Nsight Systems, Nsight Compute, GEMM 优化, FP8/FP4/INT4

机器学习与多模态: PyTorch, Hugging Face Transformers, Diffusion/DiT, 多模态学习, OpenCV

分布式系统: PyTorch Distributed, Docker, Kubernetes, Linux, Prometheus, CI/CD

编程语言: Python, C++, CUDA C++, Bash, SQL, 数据结构与算法

### 教育背景

软件工程硕士, 软件学院, 西北工业大学 (985/211) - 2024年9月 - 2027年7月

GPA: 88/100, 前 1%, 中国政府奖学金, 国际学生联合会主席 (ISU)

计算机科学与技术学士, 计算机学院, 西北工业大学 (985/211) - 2020年9月 - 2024年7月

GPA: 85/100, 前 1%, 校长奖学金, 最佳本科毕业论文奖

### 论文与荣誉

第一作者 — 《NVFP4-DiT: 面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》— IEEE TNNLS (Q1), 审稿中。

荣誉: 中国政府奖学金 (2024-2027); 校长奖学金 (2020-2024); 最佳本科毕业论文奖 (2024); 西北工业大学优秀毕业生 (2024); NWPU AI 挑战赛黑客松冠军 (2023)。